

EDUCATION

Bachelor of Engineering — Computer Science Engineering
Sathyabama Institute of Science and Technology, **CGPA: 8.3 / 10**

Aug 2024– May 2028
Chennai, Tamil Nadu

INTERNSHIP EXPERIENCE

Artificial Intelligence Intern NoviTech R&D Private Limited | Jul 2024 – Aug 2024 | Certificate ID: AIIN3859

- Built a **conversational AI chatbot** using Python with NLP-driven intent recognition and dynamic response generation, delivering end-to-end dialogue management functionality
- Designed and evaluated **deep learning classification models** using TensorFlow and scikit-learn, applying feature engineering pipelines with NumPy and Pandas to improve model accuracy and training efficiency.

PROJECTS

Predictive Maintenance System | Python · PySpark · XGBoost · Power BI

- Architected an end-to-end ML pipeline ingesting real-time industrial IoT sensor data to predict equipment failures **before occurrence**, reducing potential downtime risks proactively.
- Achieved **99.66% ROC-AUC** by engineering domain-specific features and tuning XGBoost with Bayesian optimization, outperforming baseline models by a significant margin
- Processed and transformed **large-scale time-series sensor data** using PySpark distributed computing, enabling scalable inference across multi-node environments

AWS Cloud Data Pipeline | AWS S3 · Lambda · RDS · boto3 · CloudWatch

- Designed a fully **serverless, event-driven data pipeline** on AWS — automating end-to-end flow from CSV ingestion through S3, Lambda transformation, and RDS PostgreSQL storage to scheduled reporting
- Implemented **IAM least-privilege security policies** and AWS cost optimization strategies, ensuring production-grade compliance and resource efficiency
- Configured **CloudWatch alarms and event-driven triggers** for real-time pipeline monitoring, enabling automated failure detection and recovery workflows

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, Java, Bash/Shell Scripting.
- AI & Machine Learning:** TensorFlow, PyTorch, XGBoost, MLflow, Model Evaluation.
- Data Engineering & Big Data:** Apache Spark, PySpark, Apache Kafka, Apache Airflow, ETL/ELT Pipelines.
- Cloud Platforms:** Amazon Web Services (AWS) — S3, Lambda, RDS, Google Cloud Platform
- Databases & Data Warehousing:** PostgreSQL, MySQL, MongoDB, Redis, Cassandra.
- MLOps & DevOps:** Docker, Kubernetes, Git, GitHub, CI/CD, Linux/Unix, REST APIs
- Data Visualization & Analytics:** Power BI, Microsoft Excel (Advanced), Data Analysis, Statistical Analysis.

CERTIFICATIONS

Certification	Issuer	Year
Python (Basic)	HackerRank	2025
Python for Data Science Course	Reliance Foundation	2025
AI-DevOps Engineering Foundation	Reliance Foundation	2025
Python for Data Science (Elite)	NPTEL	2025

ACHIEVEMENTS & AWARDS

Luminary Award — Excellence in Academics Sathyabama Institute of Science and Technology | 2024–2025

- Awarded institutional recognition for outstanding academic performance and dedication during the first year of BE Computer Science Engineering.